

Name: _____



Extrapolating a sequence

Task A: Other sequences

1) For each of the starting two terms, continue the sequence in three different ways. Explain the rule you have used for each way.

a) 2, 6, __, __, __, ...

c) 8, 3, __, __, __, ...

b) -1, 5, __, __, __, ...

d) 11, 13, __, __, __, ...

2) For each of these sequences, suggest what the next two terms might be. Explain your reasoning.

a) 2, 3, 8, __, __, ...

b) 31, 29, 31, __, __, ...

c) 3, 20, 190, __, __, ...

d) 2, 4, 16, __, __, ...

e) 8, 5, 4, 9, __, __, ...

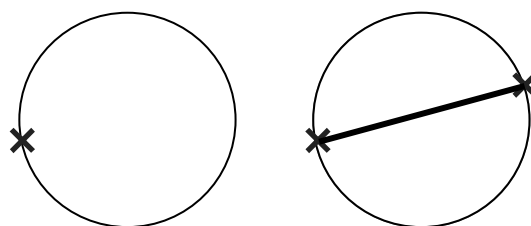


Task B: Extrapolating a sequence

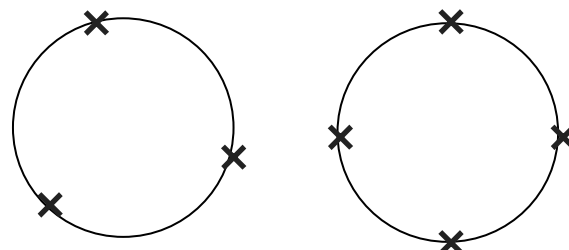
1) Fill in this table to investigate patterns found when points around a circle are joined with straight lines.

Number of points on the circle	1	2	3	4	5	6
Number of lines	0					
Number of intersections	0					
Number of regions	1					

a) Complete the table for 2 points.



b) Fill in the table for 3 points and 4 points.

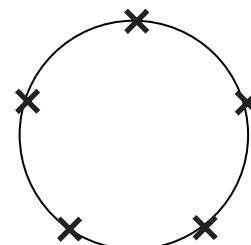


c) Describe any patterns you notice.

d) Make a prediction for what the 5th column will look like.

e) Use the circle given to fill in the 5th column.

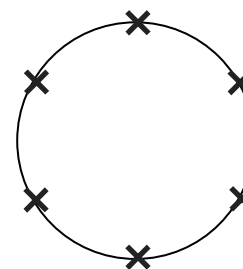
f) Were you correct, have any of your patterns changed?



g) Make a prediction for what the 6th column will look like.

h) Use the circle given to fill in the 6th column.

i) Were you correct? Write a sentence explaining what you have found:



j) Do you think you have found any patterns which will continue? Write a sentence explaining why.